CATHODE POWER | SPREADING GOODNESS WITH TRUST

User Manual

Installation | Operation | Maintenance



PV GRID CONNECTED INVERTER

CP-2300L	CP-3600L	CP-4000L	CP-4400L	CP-4600L
CP-4800L	CP-5000L	CP-5300L	CP-5500L	CP-6000L

Introduction :

For Readers

This manual is applicable for technicians of inverter installation, operation and maintenance. The readers shall be familiar with electrical knowledge.

About this Manual

Please read this manual carefully before using products. This manual shall be kept in a place which is convenient to use. Operator using this manual must be a qualified electrical engineer certified by the local electrical authority.

Copyright

This manual updates are subject to products upgrades, and there might be some unconformity between the manual description and the products, and users can acquire the latest version from us. The manual updates are not to be further notified.

Applicable Models

This manual provides the installation, operation and maintenance of PV grid-connected inverters **CP-2300L - CP-6000L**. The following models of inverter are related :

- **CP-2300L**
- **CP-3600L**
- **CP-4000L**
- **CP-4400L**
- **CP-4600L**
- **CP-4800L**
- **CP-5000L**
- **CP-5300L**
- **CP-5500L**
- **CP-6000L**

Revision

Revised Version No.	Date	Description
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CP-MAN 1.2	2024.12.16	Rev.2

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1. Safety Instruction

Symbols in this manual

To ensure the personal and property safety in using the photovoltaic inverter and the high efficiency of the product, related safe operation notices are provided in the manual, and corresponding symbols are used for emphasizing the importance. These important notices must be fully understood and followed. Symbols used in this manual are listed below to help you carefully read and use this manual.

	It means a highly potential danger which may cause a serious personal injury or death directly if this warning is neglected.
	It means a moderately potential danger which may cause a serious personal injury or death directly if this warring is neglected.
	It means a lightly potential danger which may cause a light or moderate personal injury or serious property loss if this warning is neglected.
	It means a lightly potential danger which may cause a light or moderate personal injury or serious property loss if this warning is neglected.
	It means an additional notice emphasizing or complementing the content, or providing a tip for optimizing the product operation, and further helps you solve some problems or save some time.
	It means a helpful reference or notice.

CP-2300L - CP-6000L Series inverters are designed, manufactured and tested as per international safety standards. However as an electrical and electric product, it must be installed, operated and maintained strictly according to related safety notices.

If you have any problems, please contact the nearest service center or authorized dealer. Please do NOT install or repair the product by anyone who is not qualified by local authority.

We are not responsible for any damage or loss caused by misuse or misunderstanding of the information in this manual.

Warning

Misuse or misoperation may harm :

- The personal safety of the operator or a third person
- The property safety of the inverter or any other property.

All Installation and Electrical work must only be Performed by Qualified Personal.
They have :

- Been trained specially
- Be familiar with safety requirements for electrical systems.
  The inverter must be installed and maintained by professionals in compliance with local electrical standards regulations and the requirements of local power authorities or companies.
- There is a risk of injury due to improperly handing device.
- Do not disassemble any parts of inverter which are not mentioned in installation guide. It contains no user-serviceable parts.
- Always follow the instructions contained in the manual when moving or positioning the inverter.
- Please install it in the place beyond children's reach.
- Prior to installing and maintaining the inverter, it is crucial to make certain that the inverter in not electrically connected.
- Before maintaining the inverter, disconnect the connection between the AC grid and the inverter firstly, and then disconnect the connection between the DC input and the inverter, you should wait at least 5mins after these disconnection in case of electric shock.
- All cables must be firmly attached, undamaged, properly insulated, and adequately dimensioned.
- The temperature of some parts of the inverter may exceed 60°C during operation. To avoid being burnt, do not touch the inverter during operation. Let it cool before touching it.
- Without permission, It is not allowed to open the front cover of the inverter. Users should not touch/ replace any components of the inverter except the DC/AC connectors. We will not bear any consequences caused by unauthorized actions which will lead to potential injury to people and damage to inverters.
- Static electricity may damage electronic components. Appropriate method must be adopted to prevent such damage to the inverter; otherwise the inverter may be damaged and the warranty will be annulled.
- When exposed to sunlight, the PV array will generate very high voltage which can cause electrical shock hazard. Please strictly follow the instruction we provided.
- PV modules should have an IEC61730 class A rating.
- Prohibit inserting or pulling the AC or DC terminals when the inverter is working. Or the inverter will be destroyed. Only DC connectors provided by We are permitted to use, otherwise the inverter may be damaged and the warranty will be annulled.
- Do not disassemble any parts of inverter which are not mentioned in installation guide. It contains no user-serviceable parts

- The inverter incorporates a certified internal Residual Current Device (RCD) in order to protect against possible electrocution and re hazard in case of a malfunction in the cables or the inverter. There are two trip thresholds for the RCD as required for cortication (IEC 62109-2: 2011). The default value for electrocution protection is 30 mA, and for slow rising current is 300 mA.
- If an external RCD is required by local regulations, check which type of RCD is required for relevant electric code. It recommends using a type-A RCD. The recommended RCD values is 100mA or 300mA unless a lower value is required by the specific local electric codes. When required by local regulations, the use of an RCD type B is permitted.
- The device is intended to connect to a PV generator with a capacitance limit of approx. 700nf.
- Incorrect grounding can cause physical injury, death or equipment malfunction and increase electromagnetic.
- Make sure that grounding conductor is adequately sized as required by safety regulations.
- Do not connect the ground terminals of the unit in series in case of a multiple installation. This product can cause current with a d.c component, Where a residual current operated protective (RCM) or monitoring (RCD) device is used for protection in case of direct or indirect contact, only one RCD or RCM of type B is allowed on the supply side of this product.
- Never touch either the positive or negative pole of PV connecting device. Strictly prohibit touching both of them at the same time.
- When accessing the internal circuit of solar inverter, it is very important to wait 5 minutes before operating the power circuit or demounting the electrolyte capacitors inside the device. Do not open the device beforehand since the capacitors require time to sufficiently discharge!

1.1 Before Installation :

Warning	The inverter cannot be connected to the grid unless approved by the electrical authority, and it must be installed according to the local standard and related standard for an electrical enterprise.

Notice	Please check if there is any damage on the package or the product before installing. The inverter is electrical radioactived, Please choose a suitable place for installing.

1.2 During Installation :

Danger	Keep the PV array covered and the DC circuit breaker OFF. High voltage will be generated by PV array exposed under sunshine. All the cables must be connected firmly..

Notice	The inverter must be installed by a qualified electrical engineer certified by the local authority, and the installation manual must be read through before installation. It must be installed according to the local standard and related standard for an electrical enterprise.

1.3. Operation :

Danger

- High voltage is a hazard, make sure keep the device away from children.
- Any touch with the device or terminal may cause electric shock or fire. Please follow all the safety instructions.
- A damaged device or system fault can cause electric shock. Make sure that you have checked the package and the device before installation to avoid unnecessary damage or loss.

Caution

Be aware of the hot surface while the device is running.

1.4 Repair :

Warning

Completely switch off the connection between the inverter and the grid, DC side connection. Wait for 10 minutes until the internal elements are fully discharged.

Notice

Do NOT restart the inverter before all the hazards have been removed. Please contact your local dealer and always have licensed trader do the repairing.

1.5 EMC :

EMC (Electro Magnetic Compatibility) means the resistance of a device or system against generating any Electro Magnetic interference to the environment without influencing the normal operation in the Electro Magnetic environment.

- Immunity to the own noise; Immunity to the internal electrical noise.
- Immunity to the external noise; Immunity to the external Electro Magnetic noise.
- Noise radiation level: influence of Electro Magnetic radiation to the environment.

Notice

- The Electro Magnetic radiation of the inverter is harmful for health.
- Please never stay within 20cm from a running inverter for long.

2. Product Description :

2.1 Product Applicability :

By **CP-2300L - CP-6000L** series inverter, the DC voltage generated in the PV array can be transformed into AC voltage and supplied to the grid. The PV power generating system consists of PV arrays, inverters, meters and a public grid.

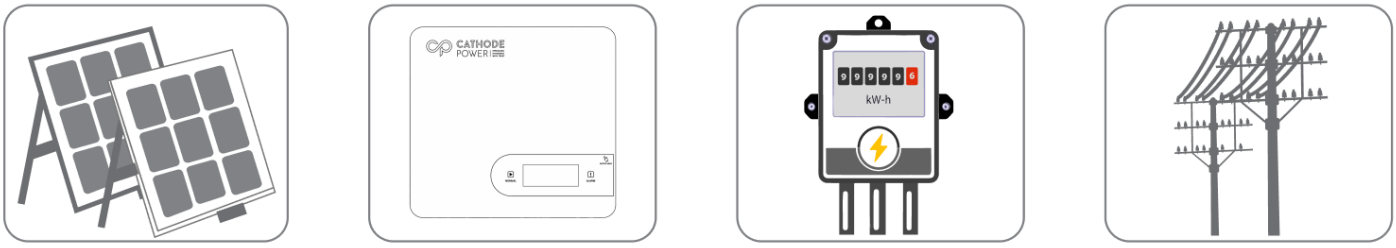


Fig. 2.1

The research, development and manufacture of **CP-2300L - CP-6000L** series are integrated with the most updated techniques and public confirmed safety regulations. However, improper operation or misuse may still cause injury or loss. Instruction and information provided in this manual must be followed all the time.

2.2 Circuit Structure :

Fig.2-2 shows the internal functional diagram of **CP-2300L - CP-6000L** series inverter. After the PV array input enters the voltage boosting circuit via the filter circuit, the input DC voltage is boosted and stabilized to BUS value for the full-bride inverter circuit, and in this process, the MPP tracker in the inverter will ensure the DC energy generated in the photovoltaic array can be used by the inverter circuit at maximum and the DC current will be conveyed into the grid.

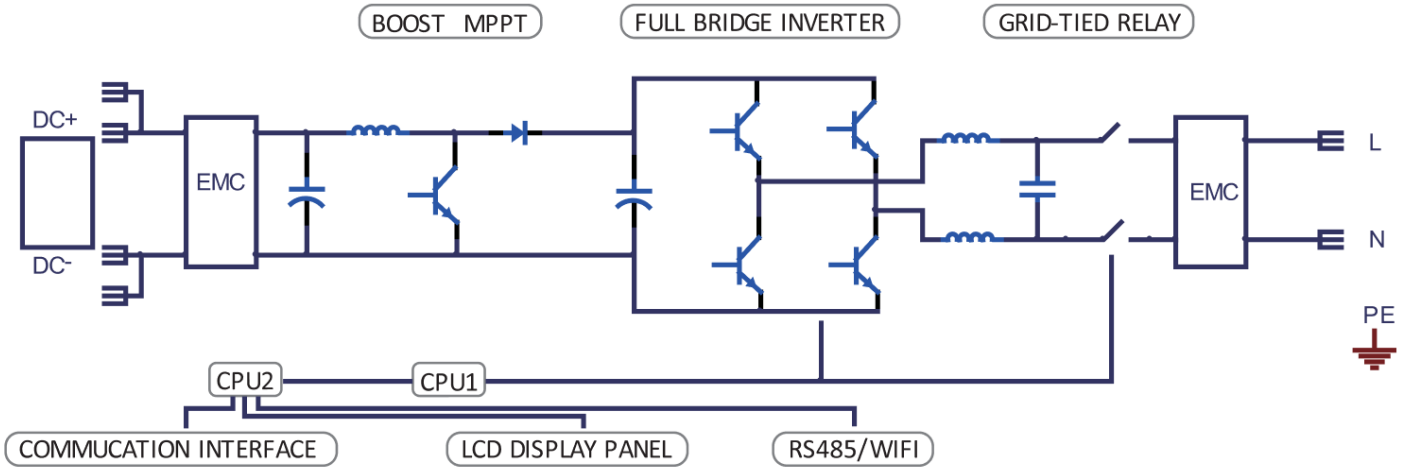


Fig. 2.2

2.3 Product Overview :

2.3.1 Electrical Connecting Part :

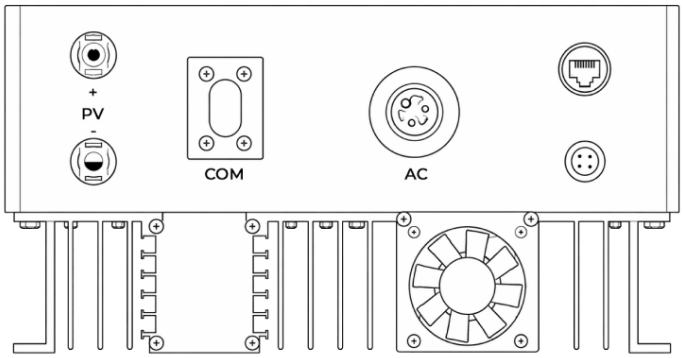


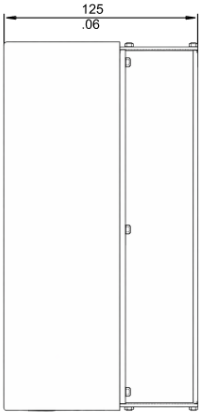
Fig. 2.3

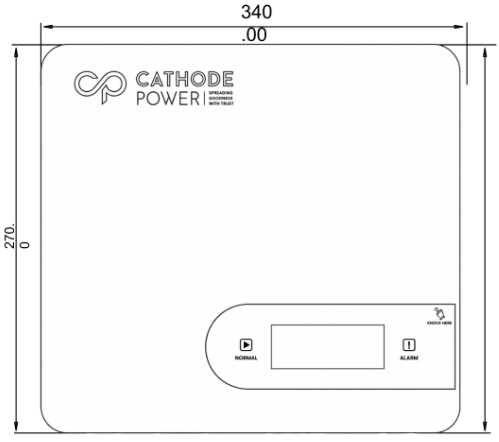
Name	Description
DC+	The positive part of terminals connecting the PV array
DC-	The negative part of terminals connecting the PV array
RS485 (1/2)	Standard communication mode
AC Output	Connected to the grid

A DC switch is recommended in consideration of safety, in some countries, it is imperative for the connection of PV array and inverter; users can select a proper type for T series inverters accordingly."

2.3.1 Dimensions and Weight :

CP-2300L - CP-6000L → 8 kg





Notice

Waterproof I/O terminal are located at the bottom of the inverter.Be cautious when handling and installing to prevent any damage.

3. Installation :

3.1 Installation Procedures :

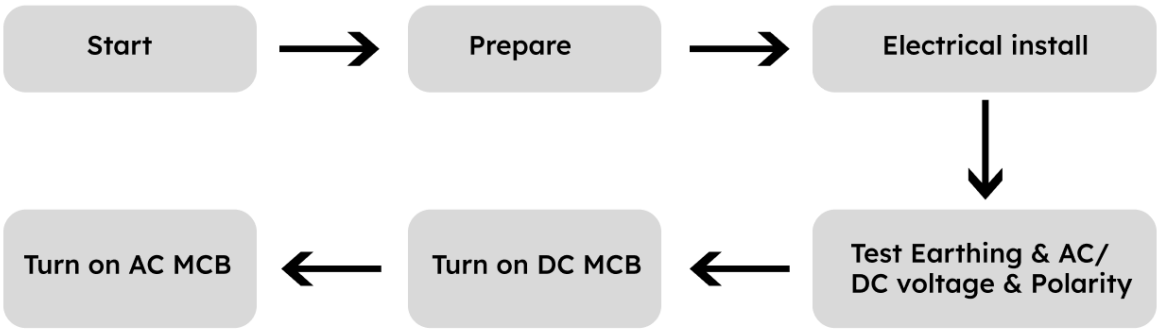


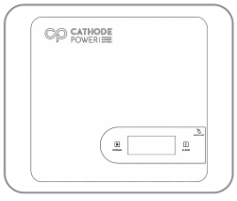
Fig. 3.1

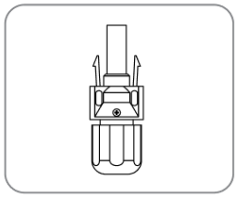
- Not be exposed to glare.
- Not in the cool air directly.
- Not near the television antenna or antenna cable.
- The ambient temperature in the range of -25℃ to +60℃. The slope of the wall should be within ±5°.
- Be sure the ventilation is good enough.
- Not higher than altitude of about 3000 m above sea level.
- Not in areas where highly flammable materials are stored.
- The wall hanging the inverter should meet conditions below :
  1. Solid brick/concrete, or strength equivalent mounting surface.
  2. Inverter must be supported or strengthened if the wall's strength isn't enough (Such as wooden wall, the wall covered by thick layer of decoration)

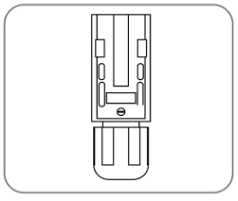
3.2 Preparation Before Installation :

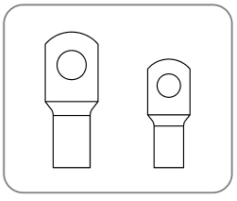
3.2.1 Unpacking and Checking :

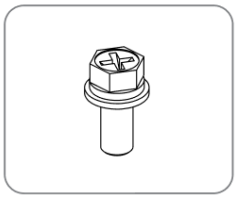
The product is carefully tested and checked before transportation, however it is still possible to be damaged during transportation, please check the device again before installation. If any damage, please contact the transportation agency or directly contact our company. Please take picture of the damage; best and efficient help will be offered.

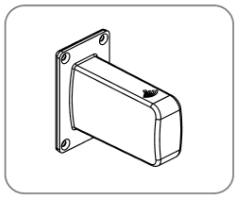

Inverter X 1

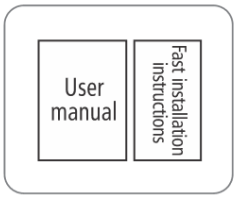

Positive DC Plug*


Negative DC Plug*

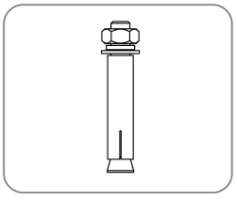

Connecting Terminal


Grounding Screw


External Wifi Module x 1



User Manual


Expansion Bolt*

3.3 Section of installation Position :

✧ The installation of the inverter should be protected under shelter from direct sunshine or any bad weather conditions as shown below:



No Direct Sunlight ✓ No Direct Rain ✓ Direct Sunlight ✘ Direct Rain ✘

✧ The inverter shall be installed in a cool & dry place with temperature from -4 C to +40 C; The environmental relative humidity is not higher than 95% and without any condensation.

✧ The inverter LCD shall be leveled with eyes and with enough space in the front for inspection.

✧ To avoid burning or electric shock, the inverter shall be installed beyond reach of children. The temperature of some parts (e.g.: the heat radiator) is high when the inverter is running.

✧ Make sure the installation position does not shake.

✧ The inverter shall be installed in a well ventilated place to ensure the normal heat radiation.

✧ The inverter shall be installed on a vertical wall, or within 5°at most if backwards to the wall.

✧ Connecting terminal is located at the bottom.

✧ No Bird Nest or Dust on Inverter Heatsink.

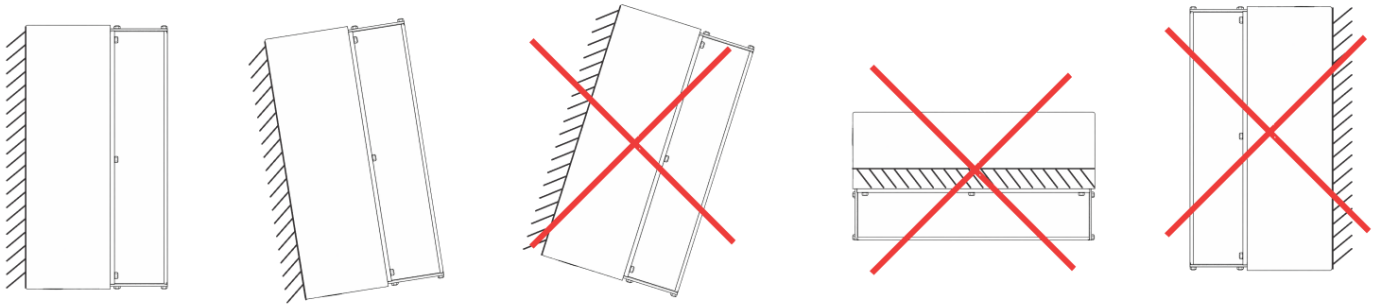
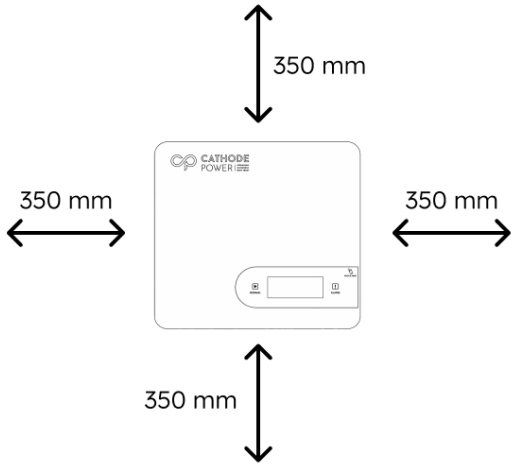


Fig. 3.3

Available Space Size :



Total Available Space Size :

Position	Min.Size
Left	350 mm
Right	350 mm
Top	350 mm
Bottom	350 mm
Front	350 mm

- CP series inverter are designed for single phase grid. Nominal grid voltage is 220/ 230/ 240 V, frequency is 50/ 60 Hz. Other technical requests should comply with the requirement of the local public grid.
- The parameters have some differences because of different environment and material. Please according to the local conditions to choose appropriate cable and micro-breaker.

3.4 Inverter Installation :

Danger	• Please check that the open circuit voltage, short circuit current and maximum power at STC of the PV array are within the capacity of the solar inverter. Danger The full load MPPT voltage range is within the 250V-480V.

✧ Take out the Inverter and fix it to the wall; then peel the three green logos on the back board and attach them to the wall through the hole. Thus the installation holes are marked.

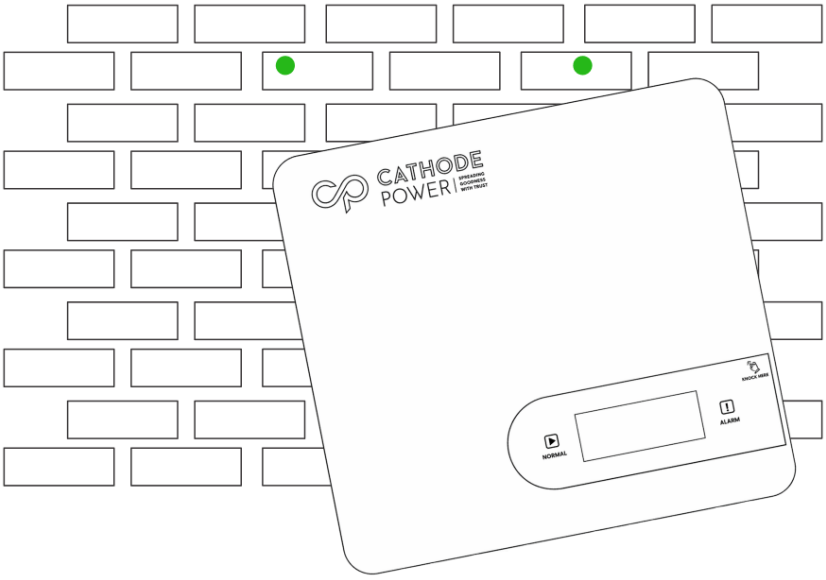


Fig. 3.3

✧ Drill holes in the marked position as per the size of expansion screws.

✧ Insert the expansion tubes into the hole, knock the tubes into the hole and make them level with the wall surface.

✧ Fix the Inverter to the wall using the tapping screws tightly.

3.5 Electrical connection :

After the inverter is correctly installed on the wall or support, the next step is the electrical connection for the inverter. Electrical connection must be performed according to related safety standards.

Danger	A misoperation electrical connection may cause personal injury or death or damage the inverter irreversibly. Wiring operation must be performed by a qualified electrical engineer.

Warning	All electrical installation must be complying with local and national electrical standards.

⚠ Warning	The inverter cannot be connected to the grid unless approved by the local electrical authority and all electrical connections are completed by a qualified electrical engineer.

⚠ Warning	Please use cables of specification recommended by us, or the system safety may be deraded.

The electrical connection for an inverter covers electrical cable connection and communication cable connection.

3.5.1 Electric and Electrical System Structure :

✧ The electric and electrical connection for the whole solar energy power generation system is shown as below :

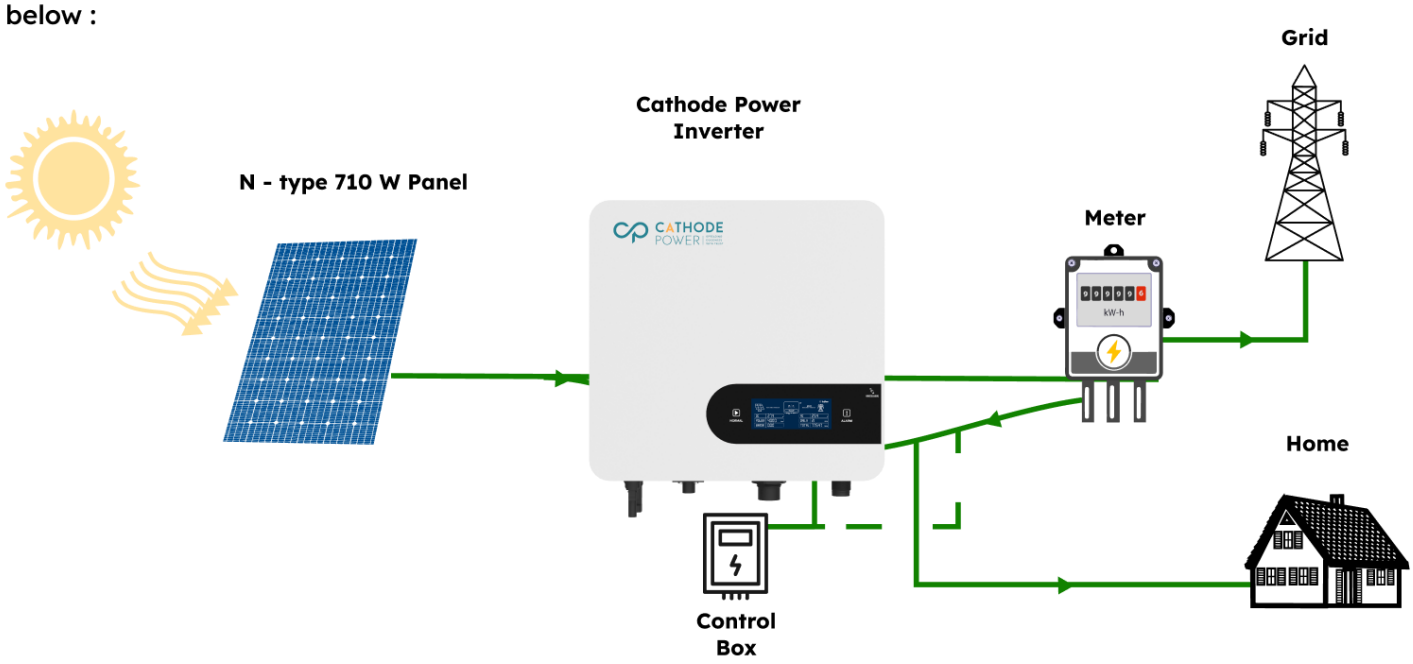


Fig. 3.5

3.4.2 Wiring Terminals and Cable Specification :

✻ DC Wiring Terminals :

1. The DC side terminals are shown in the following figure :

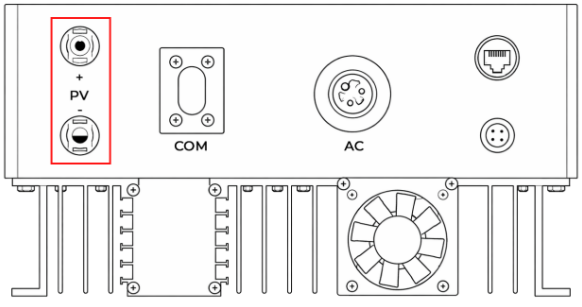


Fig. 3.5.1

2. The DC wring terminals as following :



Fig. 3.5.2



Fig. 3.5.3



Fig. 3.5.4

3. AC terminals as following :



Fig. 3.5.5



Fig. 3.5.6

3.5.3 Steps for electrical connection :

Danger	Please cover the PV array by lightproof material or switch off the DC circuit breaker before electrical connection. A dangerous voltage will be generated by the PV array exposed in the shine.
Danger	If the inverter has been electrified and tested before connection, wiring can not be performed unless the AC and DC power supplies are cut off for 10min and a multimeter shows that the DC side is tatally discharged.
Caution	When designing the PV array, make sure the maximum open circuit voltage is not higher than 550V at each series of PV group. Otherwise the inverter will be damaged irreversibly
Tip	• Different colors of cables shall be used for differentiating in assembling. For example: the positive pole is connected by a red cable and the negative by a blue cable. • In order to balance each PV series, Cable shall be with the same sectional area.
Tip	• DC cable between the inverter and the assembly shall be the special PV cable. The voltage drop from the terminal box to the inverter is about 1-2%. It is recommended the inverter is installed on the assembly support for generation to reduce the cable cost and the DC loss.
Tip	• High performance and high quality PV array shall be used. The open circuit voltage in the serial array must be lower than the maximum DC input voltage in the PV grid-tied inverter, and the working voltage of the serial array must comply with the MPPT voltage of the inverter.
Notice	• Protect the LCD panel during wiring. • Protect the LCD panel and other elements from scratch or damage by cables or tools.

3.5.3.1 Steps for DC Connection :

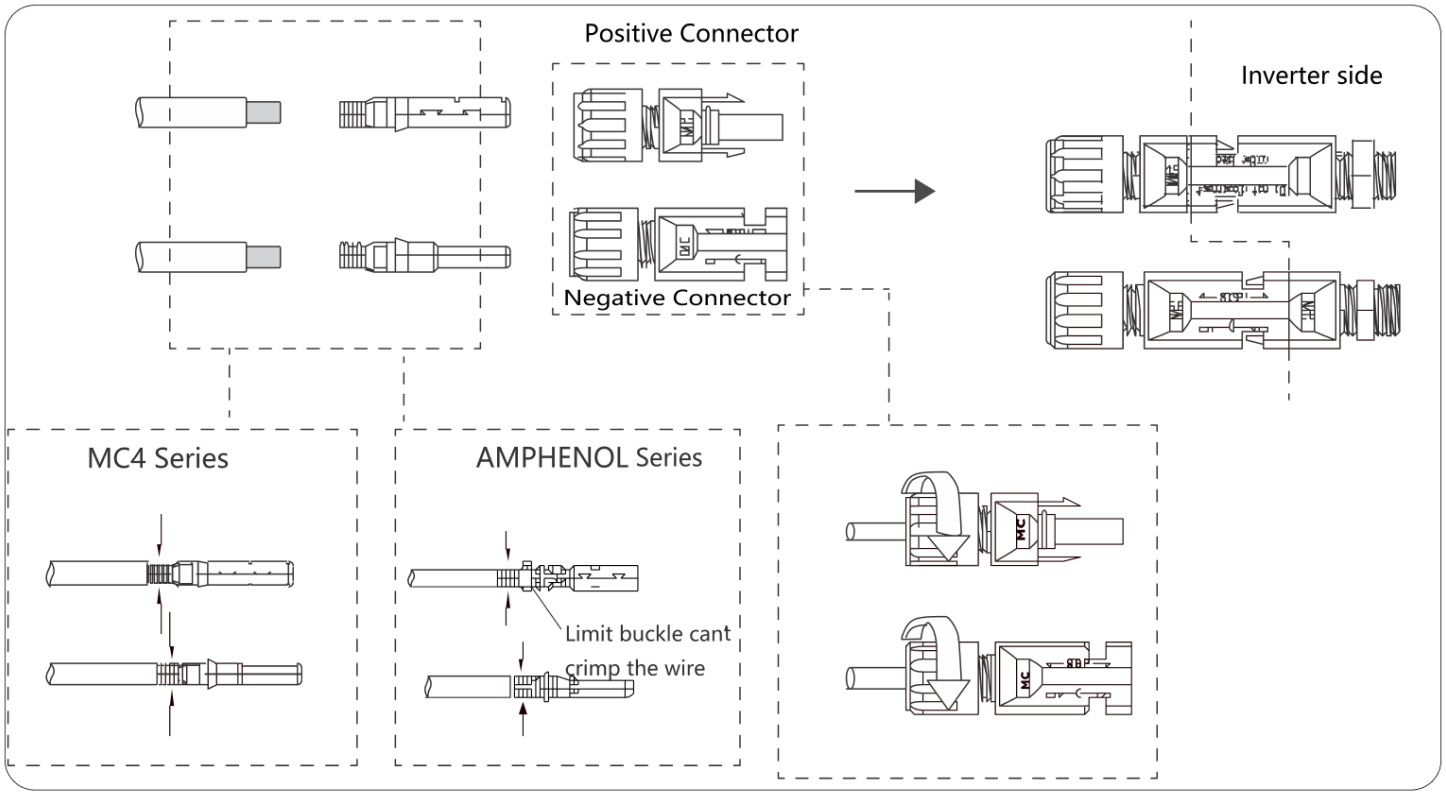


Fig. 3.5.7

3.5.3.2 Steps for AC Connection :

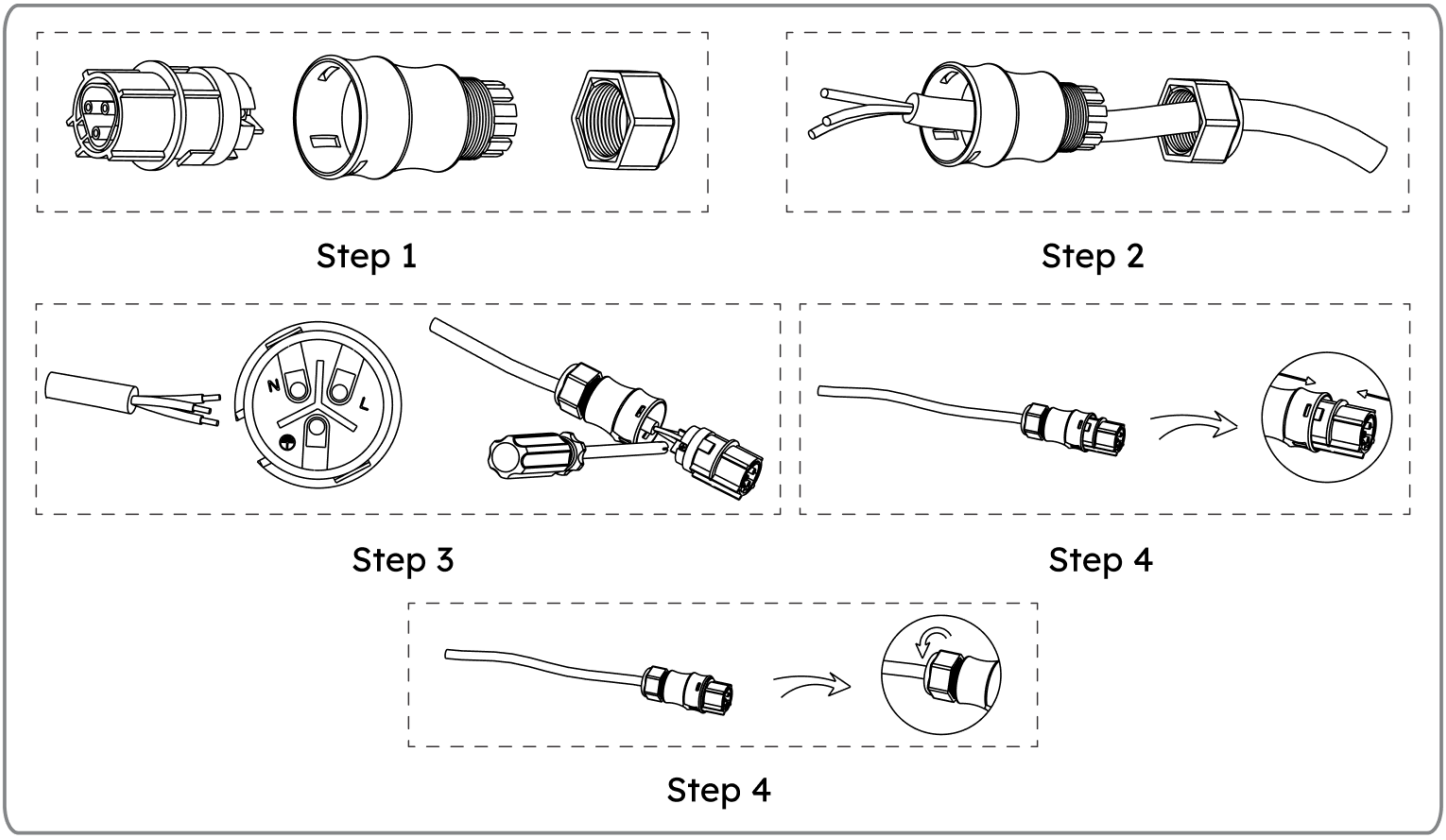


Fig. 3.5.8

3.5.3.3 DC/AC Terminal Connection :

- Make Sure that the DC/AC Breaker is Switched Off.
- Insert the DC+/- wiring terminal into corresponding DC+/- terminal.

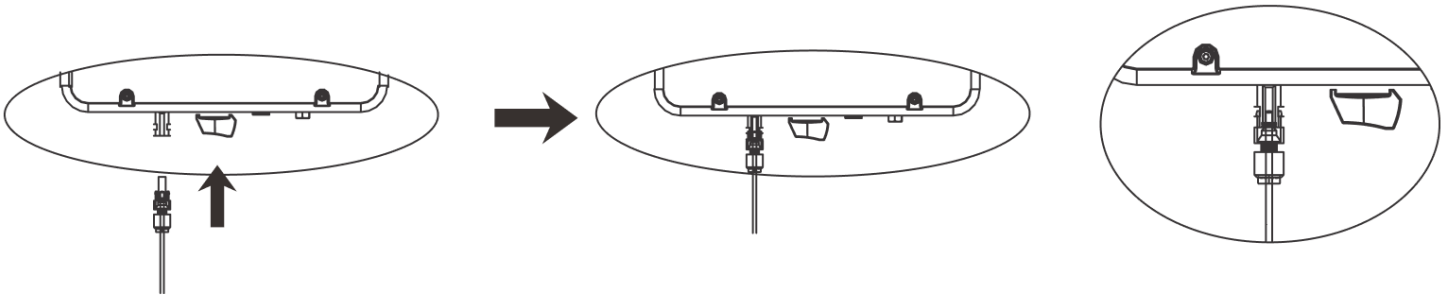


Fig. 3.5.9

- Insert the AC wiring Terminal into the AC terminal.

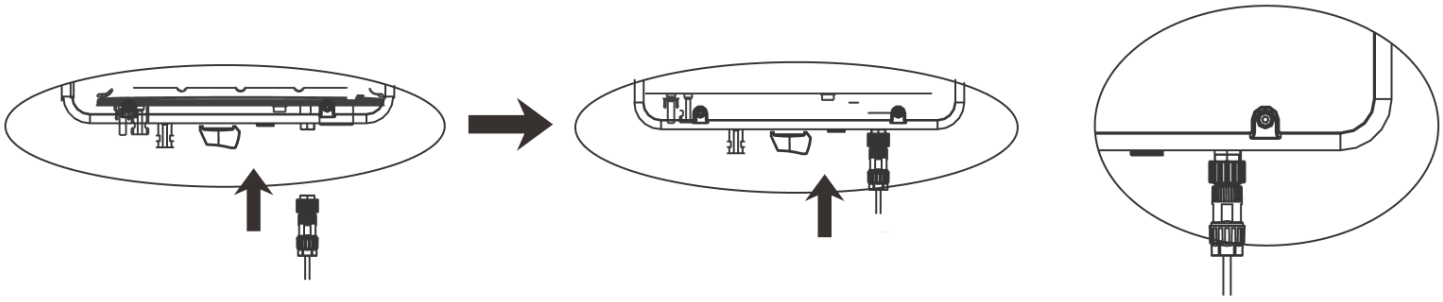


Fig. 3.5.10

- Switch on the DC breaker.
- Switch on the AC breaker.

3.4 Earthing Requirement :

⚠ Caution	The inverter is a device with no transformer, neither the positive nor the negative pole of the PV assembly DC side can be earthed, or the inverter will not work normally, or even be damaged irreversibly.

- In series PV power generating system, all device shells, assembly supports and the GND terminal of the inverter must be safety and reliably earthed.
- The Earthing to Normal voltage must be less than 3V

4. Trial Operation :

4.1 Check Before Operation :

Warning	Necessary safety check must be performed before the inverter is electrified for trial operation!

4.1.1 Check for Reliability of Mechanical Installation :

- Check if the inverter is firmly installed and if all bolts are reliably tightened. For an inverter installed on a metal supporter, make sure each bolt is tightened and support has enough load bearing capacity.

4.1.2 Check for connecting cables :

- Check if all cables in the system are firmly connected without any missed or wrong connection, and especially check if all positive and negative poles are correct. If a DC switch is equipped on the inverter, the DC switch shall be turn to the "OFF" state.

4.1.2 Electrical check :

- Make sure the DC input voltage of the inverter is lower than 550V (with the temperature decrease of the PV array, the open circuit voltage will be increased, so a residual voltage at the low temperature must be put into consideration).

4.2 Electrify the Inverter :

- The inverter can be started up as per the following steps after all testing and checking steps are performed.
- Switch on the DC breaker, and then the AC breaker, if an inverter is with a DC switch, turn it to the "ON" state.

- If the inverter is not normally started up, please refer to Part 5 "Troubleshooting and maintenance".

5. Troubleshooting and Maintenance :

5.1 Troubleshooting :

- This part introduces the common fault and solving steps, provides troubleshooting method and skills to the user, and helps the user identify and solve some common fault of the inverter.

Tip	• Faults mentioned in this part means the inverter enters the protection state or fault state.

- The protection state is mostly caused by some external reasons, the inverter output is not matched with the grid, the grid is power off or the system is not correctly earthed, and thus the inverter acts to ensure the own safety. The inverter itself has no defect and can recover from the protection state if the system is correctly designed and installed.
- Sometimes the inverter enters the protection state by the own defect and can be recovered by resetting or repairing.
- The fault state is caused by incorrect design of system, e.g: the DC voltage is too high, and thus the inverter acts to protect the system safety. The inverter can be recovered from some faults automatically or by resetting, and some faults are caused by the own defect of the inverter, and in this case the inverter cannot be recovered and shall be repaired.
- Correctly identifying the fault state of the inverter is helpful for solving the fault, so please check as per the following steps :

1. Check if the status displayed on the panel is a protection state or fault state, and record all error codes.
2. Try the solution listed in Tab.5-1 and 5-2 and meanwhile check as per the following steps.
   - If the voltage at the AC/DC side is normal and if the breaker is switched on?
   - If the inverter is installed in a clean, dry and well ventilated place?
   - If the cable sectional area satisfies the requirement?
   - If the connecting cables is too long? •If all cables are firmly connected?
   - If wiring is reasonable?
   - If the safety standard setting is correct?
3. If you meet any problem that you cannot solve by yourself, please contact the client service man for the optimal solution, please provide the detailed system installation information, inverter model, serial number and fault information.

Tab 5.1 Protection code and solution ：

Displayed code	Protection Description	Recommended solution
P001	The leakage current is too high	1. Cut off the DC breaker and check if there is abnormity in the AC side. 2. After the fault is removed, close the DC breaker and restart the inverter. 3. If the fault occurs repeatedly, please ask for professional service.
P002	The voltage at the grid side is too high	1. Check if the inverter is normally connected to the grid. 2. Check if the grid voltage and frequency are complying with the output scope of the inverter. 3. If the fault above is removed but the inverter can still not work normally. Please ask for professional service.
P003	The voltage at the grid side is too low	
P004	The frequency at the grid side is too high	
P005	The frequency at the grid side is too low	
P006	Islanding protection	
P007	The DC voltage is too low	1. The inverter will recover automatically when the light is sufficient.
P008	Output short circuit	1. Check if the inverter is normally connected to the grid. 2. Check if the grid voltage and frequency are complying with the output scope of the inverter. 3. If the fault above is removed but the inverter can still not work normally, please ask for professional service.
P009	Output current DC offset is too high	
P010	Internal synchro communication fail	1. Switch off the DC breaker and wait 1 minute. 2. Switch on the DC breaker and check if the inverter can auto reboot. 3. If the fault occurs repeatedly, please ask for professional service.

Tab 5.2 Fault code and Solution ：

Displayed code	Fault Description	Recommended Solution
E001	The Insulation Resistance is low	1. Switch off the breaker. 2. Check if the resistance between the +/- poles of the PV assembly and the ground is larger than 1MΩ. 3. If the fault above is remove but the inverter can still not work normally, please ask for professional service.
E002	The DC voltage is too high	1. Switch off the DC breaker. 2. Check if DC voltage is higher than (550V for 2.3 - 4.0 kw) & (550V for 4.4 to 6.0 kw) 3. If the fault above is removed but the inverter can still not work normally, please ask for professional service.
E003	The DC input current is too high	1. Switch off the DC breaker. 2. Check if the system is designed incorrectly(DC power too high). 3. If the fault above is removed but the inverter can still not work normally, please ask for professional service.

5.2 Daily Maintenance ：

It is unnecessary to maintain the inverter in normal conditions but it is necessary to make sure the inverter is not covered by dust.

- Cleaning :

The inverter can be cleaned by electric compressed air blower, dry soft cloth or soft brush. Please never clean the inverter by any water, corrosive chemical agent or cleaning agent.

- Heat Radiation :

To protect the normal operation and prolong the service life of the inverter, the inverter must be installed in a well ventilated space without any air barrier surrounding the heat radiator at the back of the inverter, and dust or snow must be removed timely if any.

6. Uninstallation :

The inverter shall be treated as per the following steps if it is to be replaced or out of service.

6.1 Uninstalling Steps :

- Switch off the AC breaker.
- Switch off the DC breaker.
- Wait for 10min till the inverter is fully discharged.
- Check by a multimeter and make sure neither the AC side nor the DC side is electrified.
- Press the right position of the terminal as Fig. 6-1, When you hear a "click", then you can pull out the positive and negative DC terminal adaptor

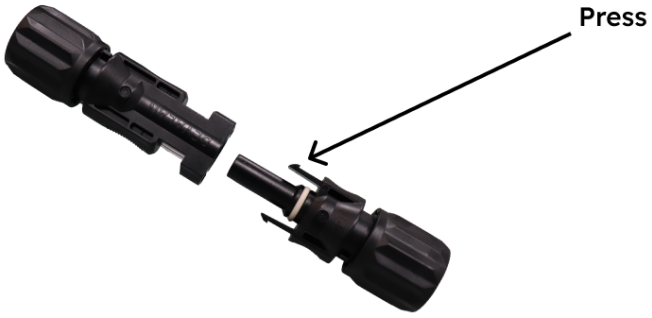


Fig. 6.1

6.2 Packing :

Please pack the inverter with the original package if possible. If the original package is not available, please pack with a carton satisfying the following requirement.

- Firm enough for 45kg.
- Easy for handing.
- Fully closable

6.3 Storing :

- The inverter must be stored in a dry and clean place of -30 C-+80 C.

6.4 Solution at the Termination of Service Period

- At the termination of inverter service period, please send the inverter and package material to the designated place for electronic waste treatment. Contact the related authority for treating and recovering if necessary. No inverter or any part can be disposed at random.

7. Technical Parameters :

TECHNICAL DATA
(CP-2300L TO CP-4400L)

MODEL NO.	CP-2300L	CP-3000L	CP-3400L	CP-3600L	CP-4000L	CP-4400L
DC Side | Input Parameters						
Max. Allowable DC Voltage (VDC) | Max. DC Voltage (VDC)	275/450V	320/450V	320/450V	320/450V	385/450V	450/450V
Min. System Start | Shut Down Voltage (VDC)	45V	45V	45V	45V	45V	45V
Max. Input Current (A)	18A	18A	18A	18A	18A	18A
Number of MPP Tracker | Total Nos. Of String	1/1	1/1	1/1	1/1	1/1	1/1
Input Terminal Type	MC4					
AC Side | Output Parameters						
Rated Output Power (W)	2300W	3000W	3400W	3600W	4000W	4400W
Maximum Output Power (VA)	2300VA	3000VA	3400VA	3600VA	4000VA	4400VA
Nominal Output Voltage/Range (V)	L/N/PE,240/140-310Vac					
AC Grid Frequency/Range (Hz)	50Hz / 45Hz-55Hz					
Max. Output Current (A)	9.5A	12.5A	14.2A	15A	16.8A	18.3A
AC Connection (With PE)	Single Phase					
Current Distortion (THDi)	<1.5%	<1.5%	<1.5%	<1.5%	<1.5%	<1.5%
Power Factor	>0.99 (Adjustable From 0.8 Leading To 0.8 Lagging)					
Efficiency						
Maximum Conversion Efficiency	97.3%	97.3%	97.4%	97.5%	97.8%	97.8%
European Efficiency	97%	97.1%	97.2%	97.2%	97.3%	97.3%
MPPT Efficiency	99.9%	99.9%	99.9%	99.9%	99.9%	99.9%
Safety & Protection						
DC Reverse-Polarity Protection	YES					
Anti-Islanding / Overvoltage Protection	YES					
Shortcut Circuit Protection	YES					
Leakage Current Protection	YES					
Grid Monitoring / Ground Fault Monitoring	YES					
DC/AC Side SPD (Thermally Protected)	YES | TYPE I + II					
General Parameters						
Dimension (L/W/H) (Mm)	340/270/127					
Weight (Kg)	7KG					
Embedded DC Switch	Optional					
Zero Export Function	Optional					
Night Power Consumption (W)	<0.2					
Isolation Type	Transformerless					
Protection Degree	IP66 According To IEC60529					
Operation Temperature (°C)	-25 - +60					
Cooling Concept	Natural Convection + Fan Cooling (12000 RPM)					
Operating Altitude (M)	<2000m Without Power Derating					
Acoustic Noise Level (DB)	<35 DB					
Display	Graphic LCD					
Communication Interface	Standard WIFI; RS485 (Optional)					
Warranty	10 Years ; 25 Years For Optional					

7. Technical Parameters :

TECHNICAL DATA
(CP-4600L TO CP-6000L)

MODEL NO.	CP-4600L	CP-4800L	CP-5000L	CP-5300L	CP-5500L	CP-6000L
DC Side | Input Parameters						
Max. Allowable DC Voltage (VDC) | Max. DC Voltage (VDC)	420/550V	500/550V	550/550V	550/550V	550/550V	550/550V
Min. System Start | Shut Down Voltage (VDC)	45V	45V	45V	45V	45V	45V
Max. Input Current (A)	22A	22A	22A	22A	22A	22A
Number of MPP Tracker | Total Nos. Of String	1/1	1/1	1/1	1/1	1/1	1/1
Input Terminal Type	MC4					
AC Side | Output Parameters						
Rated Output Power (W)	4600W	4800W	5000W	5300W	5500W	6000W
Maximum Output Power (VA)	4600VA	4800VA	5000VA	5300VA	5500VA	6000VA
Nominal Output Voltage/Range (V)	L/N/PE,240/140-310Vac					
AC Grid Frequency/Range (Hz)	50Hz / 45Hz - 55Hz					
Max. Output Current (A)	19.1A	20A	20.8A	22A	23A	25A
AC Connection (With PE)	Single Phase					
Current Distortion (THDi)	<1.5%	<1.5%	<1.5%	<1.5%	<1.5%	<1.5%
Power Factor	>0.99 (Adjustable From 0.8 Leading To 0.8 Lagging)					
Efficiency						
Maximum Conversion Efficiency	97.4%	97.5%	97.5%	97.5%	97.6%	97.6%
European Efficiency	97.2%	97.3%	97.3%	97.4%	97.4%	97.4%
MPPT Efficiency	99.9%	99.9%	99.9%	99.9%	99.9%	99.9%
Safety & Protection						
DC Reverse-Polarity Protection	YES					
Anti-Islanding / Overvoltage Protection	YES					
Shortcut Circuit Protection	YES					
Leakage Current Protection	YES					
Grid Monitoring / Ground Fault Monitoring	YES					
DC/AC Side SPD (Thermally Protected)	YES | TYPE 1 + 11					
General Parameters						
Dimension (L/W/H) (Mm)	340/270/127					
Weight (Kg)	8KG					
Embedded DC Switch	Optional					
Zero Export Function	Optional					
Night Power Consumption (W)	<0.2					
Isolation Type	Transformerless					
Protection Degree	IP66 According To IEC60529					
Operation Temperature (°C)	-25 - +60					
Cooling Concept	Natural Convection + Fan Cooling (12000 RPM)					
Operating Altitude (M)	<2000m Without Power Derating					
Acoustic Noise Level (DB)	<35 DB					
Display	Graphic LCD					
Communication Interface	Standard WIFI; RS485 (Optional)					
Warranty	10 Years ; 25 Years For Optional					

8. Quality Assurance :

8.1 Warranty Policy :

For our customers we provide 5 years standard warranty from the date of installation, you can also upgrade your standard warranty to 10 years. Please keep the purchasing invoice and trade mark clear for warranty. For more information, please contact your local dealer.

8.2 Exception Clause :

- Use of undersigned purpose.
- The inverter is damaged by any force majeure (electric shock, fire accident, earthquake or seaquake, ect).
- Operating beyond safety regulations.
- The "warranty card" has not been sent to the distributor or our company.
- Without the consent of our company to change equipment or replace parts.
- Incorrect installation, debugging and use methods.
- Failure to follow the product user manual, installation manual and maintenance guidelines.
- Improper use or misuse of the device.
- The product maintenance process does not follow relevant standards.
- Any damage to inverter due to flood or enter of water into inverter due to water logging go reinstallation /operation error.
- Inverter is used or operated against any items in local policy.
- Warranty void if Bird Nest or Dust on Inverter .
- Inverter's warranty is expired if doesn't buy extended warranty.
- Warranty void if custom Opening the Inverter Cover or breaking the product seal.
- Warranty void if not connected to a specific type of solar panel.
- Regular cleaning is not performed according to a strict schedule Warranty void.
- If the warranty period is unreasonably short compared to industry standards, it might be considered invalid. For example, if the standard warranty for solar inverters is typically 5-10 years, a warranty period of only 1 year might be deemed unfair.
- Any clause that violates local, state, or federal laws is invalid. For instance, a clause that limits the consumer’s legal rights beyond what is allowed by consumer protection laws would be considered invalid.
- Ensure that all terms are reasonable and provide meaningful coverage in line with industry standards.
- Obtain the communication protocol from other illegal channels.
- Any damage to inverter due to hazardous / chemical/ pollutant environment is not covered in warranty.
- Invalidated by integration with third-party monitoring devices.
- Warranty void if customer modifies their electrical wiring without manufacturer approval

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Cathode Power Pvt. Ltd..

Corporate Office: SF 201-203, Fortune Platinum, Nr. Tulsidham Circle, Manjalpur, Vadodara, Gujarat-390011

Factory Office: R.S. No.229, Sugar Factory, Ta. Karjan, Opp. Gandhara Gandhara, Gujarat-391210

Contact: +91 98002 68002

Email: hello@cathodepower.com

Website: www.cathodepower.com